



Read/Write Structure From/to a File in C

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For writing in the file, it is easy to write string or int to file using **fprintf** and **putc**, but you might have faced difficulty when writing contents of the struct. **fwrite** and **fread** make tasks easier when you want to write and read blocks of data.

Writing Structure to a File using fwrite

We can use `fwrite()` function to easily write a structure in a file. `fwrite()` function writes the to the file stream in the form of binary data block.

Syntax of fwrite()

```
size_t fwrite(const void *ptr, size_t size, size_t nmemb, FILE *stream)
```

Parameters

- **ptr**: pointer to the block of memory to be written.
- **size**: the size of each element to be written (in bytes).
- **nmemb**: umber of elements.
- **stream**: FILE pointer to the output file stream.

Return Value

- Number of objects written.

Example

C

```
// C program for writing  
// struct to file
```

```
#include <stdio.h>
#include <stdlib.h>

// a struct to be read and written
struct person {
    int id;
    char fname[20];
    char lname[20];
};

int main()
{
    FILE* outfile;

    // open file for writing
    outfile = fopen("person.bin", "wb");
    if (outfile == NULL) {
        fprintf(stderr, "\nError opened file\n");
        exit(1);
    }

    struct person input1 = { 1, "rohan", "sharma" };

    // write struct to file
    int flag = 0;
    flag = fwrite(&input1, sizeof(struct person), 1,
                  outfile);
    if (flag) {
        printf("Contents of the structure written "
              "successfully");
    }
    else
        printf("Error Writing to File!");

    // close file
    fclose(outfile);

    return 0;
}
```

Output

Contents of the structure written successfully

Reading Structure from a File using fread

We can easily read structure from a file using `fread()` function. This function reads a block of memory from the given stream.

Syntax of `fread()`

```
size_t fread(void *ptr, size_t size, size_t nmemb, FILE *stream)
```

Parameters

- **ptr**: pointer to the block of memory to read.
- **size**: the size of each element to read(in bytes).
- **nmemb**: number of elements.
- **stream**: FILE pointer to the input file stream.

Return Value

- Number of objects written.

Example

C

```
// C program for reading
// struct from a file
#include <stdio.h>
#include <stdlib.h>

// struct person with 3 fields
struct person {
    int id;
    char fname[20];
    char lname[20];
};

// Driver program
int main()
{
    FILE* infile;

    // Open person.dat for reading
    infile = fopen("person1.dat", "wb+");
```

```
if (infile == NULL) {
    fprintf(stderr, "\nError opening file\n");
    exit(1);
}

struct person write_struct = { 1, "Rohan", "Sharma" };

// writing to file
fwrite(&write_struct, sizeof(write_struct), 1, infile);

struct person read_struct;

// setting pointer to start of the file
rewind(infile);

// reading to read_struct
fread(&read_struct, sizeof(read_struct), 1, infile);

printf("Name: %s %s \nID: %d", read_struct.fname,
       read_struct.lname, read_struct.id);

// close file
fclose(infile);

return 0;
}
```

Output

Name: Rohan Sharma
ID: 1

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#include <stdio.h> int main() { int i = 1024; for (; i > >= 1) printf("GeeksQuiz"); return 0; } How many times will GeeksQuiz be...

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C/C++ Code #include #define PRINT(i, limit) do { \ if (i++ How many times GeeksQuiz is printed in the above program ? (A)1 (B)3 (C)4 (D)Compile-time err...

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